

EDUCATION

- **Indian Institute of Technology (BHU) Varanasi** Varanasi, India
B. Tech. in Computer Science and Engineering - CGPA: 9.75/10.00 - Department Rank 1 July 2014 - May 2018

RESEARCH EXPERIENCE

- **Microsoft Research India** Bangalore, India
Research Fellow July 2018 - Present
 - Advised by Dr. Muthian Sivathanu and Dr. Ramchandran Ramjee.
 - **Gandiva** is a Cluster Scheduler/Resource Manager for Deep Learning Training jobs utilizing CPU Scheduling primitives like time-slicing, migration, and profiling to schedule GPUs efficiently. I redesigned and implemented the entire cluster manager using **Kubernetes** to manage job containers and wrote the scheduling framework in **Scala** using the **Akka Actors** library for concurrency and **gRPC** for performing RPCs.
 - **GPU Proxy** is a mechanism inspired from **CRCUDA** to enable suspend/resume, and migration of Deep Learning jobs written in **PyTorch** and **TensorFlow**. I implemented **Horovod** and **NCCL** support to enable suspend/resume and migration of distributed jobs.
 - **Prana** is a set of Proportionally Fair Scheduling policies for Gandiva based on the concept of using job migration to balance the ticket-adjusted load across servers and then performing intra-server proportional share to achieve cluster-level proportional share. It transparently handles GPU heterogeneity using profiling and a novel GPU trading strategy to improve cluster efficiency while preserving fairness. I was involved in the design of the scheduling algorithm and policies and their implementation in the aforementioned framework.

PUBLICATIONS

- **S. CHAUDHARY**, R. RAMJEE, M. SIVATHANU, N. KWATRA, and S. VISWANATHA. *Balancing Efficiency and Fairness in Heterogeneous GPU Clusters for Deep Learning*. 2020. In Proceedings of the Fifteenth European Conference on Computer Systems (EuroSys '20). <https://dl.acm.org/doi/abs/10.1145/3342195.3387555>

INTERNSHIP

- **Goldman Sachs** Bangalore, India
Summer Analyst May 2017 - July 2017
 - **DbBench** is a distributed Database Benchmarking system for **SecDB**, a proprietary object database, used to make critical decisions about the future of the database system. It can emulate the real-life workload of hundreds of clients accessing a single database endpoint and report aggregate metrics like throughput and transactions per second. I designed and implemented the entire system in **Scala** using **Akka Actors** library for concurrency and **Netty** for composable high-throughput, low-latency asynchronous socket IO. I also implemented a **YAML** based configuration system and an interpreted Turing Complete programming language to write performance tests in.

SELECTED PROJECTS

- **LTDP for Viterbi Algorithm** <https://github.com/5hubh4m/ltdp-viterbi-algorithm> Applied the Linear Tropical Dynamic Programming model to the Viterbi algorithm to parallelise the otherwise linear algorithm using the theorem of Rank Convergence. Implemented it using **OpenMPI** in conjunction with **OpenMP** in **C++**. Investigated the improvement over vanilla LTDP by redistributing data to processors after each principal iteration to prevent otherwise redundant computation of converged rows.
- **Scheme Interpreter in Haskell** <https://github.com/5hubh4m/simple-scheme> Wrote an Interpreter for a subset of Lisp (the R5RS standard) in **Haskell**, a pure functional programming language. Implemented an REPL, a parser, variables, user-defined functions using Functional Programming concepts like maps, folds, composition, monads, do-notation, currying, pattern-matching, etc.

- **Relational Database and Relational Algebra Interpreter** https://github.com/5hubh4m/simple_ra
Implemented a persistent Relational Database in C++. Hand-wrote a top-down Recursive Descent parser for interpreting a custom $LR - 1$ grammar for Relational Algebra supporting operations like σ , ρ , Π , $\times$, $\cup$, $\bowtie$, $-$, $\div$, and aggregate operations.

TEACHING

- **Introduction to Computer Programming** IIT (BHU) Varanasi
TA and Tutor *July 2016 - May 2018*
 - Responsibilities included delivering instruction during practical sessions on programming fundamentals in C, providing support and resolving queries during tutorial sessions, and curating and grading assignments for a class of about 60 undergraduates.
- **Artificial Intelligence** IIT (BHU) Varanasi
TA *Jan 2018 - May 2018*
 - Responsibilities included delivering instruction during practical sessions on fundamentals of Artificial Intelligence (Search, Logic, and Machine Learning), curating assignments, and building and managing a `git` based encrypted automatic submission system for their evaluation for a class of about 80 undergraduates.

ACHIEVEMENTS

- **IIT (BHU) Varanasi Medal** for being Department Rank 1 in Computer Science and Engineering B. Tech Class of 2014-18.
- **SRFP 2016** Awarded the Summer Research Fellowship by the Indian Academy of Sciences, Bangalore.
- **Runner Up** at Microsoft's Code.Fun.Do 2016 24-Hrs Hackathon.
- **Winner** of Parliamentary Debate Competition at AMU Literary Festival, AMU Aligarh, 2015.

POSITIONS OF RESPONSIBILITY

- **Event Coordinator and Problem Curator** for Capture the Flag event at Technex 2017, and CodeFest 2017, with the participation of over 500 students from India and abroad.
- **Jt. Secretary** of the Literary Club of IIT (BHU) Varanasi for the session 2015-16.
- **Web Development** Worked on the Website Development and Technical Committee of Kashiyaatra 2016, FMC Weekend 2015, and IITBHUMUN 2015.

SKILLS AND AREAS OF INTEREST

- **Languages** C++, C, Scala, Java, Python, Haskell, C#, ECMAScript
- **Technologies** POSIX, Web Development, Android Development, Docker, Kubernetes, L^AT_EX, MATLAB
- **Areas of Interest** Parallel, Distributed, and High Performance System

RELEVANT COURSEWORK

CSO204 - Operating Systems	Compiler Design - CSE425
CSE202 - Artificial Intelligence	Computer Networks - CSE433
CSO324 - Operation Research	Information Security - CSE530
CSE361 - Database Management Systems	Network Security - CSE537
CSE371 - Parallel Computing	Optimisation Techniques - MA526
CS231n - Convolutional Neural Networks for Visual Recognition (Stanford University)	